Networked Belief Formation and Portfolio Allocation

Endogenous Price Formation Under Topological Constraints

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Abstract

Portfolio weights are projections of belief. In the Markowitz framework, deviations from equal-weighting encode implicit claims about expected return. This paper treats those claims not as primitives but as outputs of a social process: agents generate evidence, exchange it across a network, and update credences under Bayesian or Jeffrey rules.

We show that network topology governs both the speed and reliability with which markets move from ignorance to conviction; that mistrust between agents is structurally equivalent to increased effective risk aversion; and that price formation itself can become endogenous when exogenous evidence is weak, contested, or absent. In such regimes, prices reflect the internal dynamics of the belief network rather than the underlying return-generating process.

An endogenous market model calibrated to FTSE data (2008–2023) illustrates three distinct regimes: hub-dominated fragility during the financial crisis, polarised subnetworks during the Brexit period, and dense, self-referential dynamics during COVID-19. These correspond to different configurations of network topology, mistrust, and effective risk aversion.

The central implication is that markets are not passive aggregators of information but dynamical epistemic systems. Prices are observable fixed points of a networked belief process whose structure evolves over time.

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The Markowitz Objective as Belief-to-Allocation Map

The standard mean-variance utility function is

$$U = \mathbf{w}^\top \mathbf{E} - \lambda\, \mathbf{w}^\top \Sigma\, \mathbf{w}, \tag{1}$$

where $\mathbf{w}$ is the portfolio weight vector, $\mathbf{E}$ the expected return vector, Σ the covariance matrix, and λ a scalar encoding risk aversion.

In the two-asset case—asset A with known return 0.5, asset B with true return $0.5 + \varepsilon$—the optimal weight in B is

$$w_B^* = \frac{1}{2} + \frac{p_i\, \varepsilon}{2\lambda\sigma^2}, \tag{2}$$

where p_i is the agent's credence that $\varepsilon > 0$.

Three boundary cases fix the interpretation:

- $p_i = 0.5$: the tilt vanishes, recovering equal-weight. This is maximum ignorance.
- $p_i \to 1$: the weight shifts maximally toward B, constrained by λ.
- $p_i < 0.5$: the agent tilts toward A, actively backing the wrong asset.

Every departure from $w_i = 1/N$ is a knowledge claim, scaled by willingness to act on uncertain belief.

Belief Formation in Networks

The credence p_i is not given. It is constructed through iterated interaction. Agents run experiments (binomial trials with success probability $0.5 + \varepsilon$ under B), share results with network neighbours, and update.

Under Bayes' rule, agents treat received evidence at face value, computing posterior odds from the likelihood ratio. Under Jeffrey's rule with a homophily assumption [3], agents discount evidence from neighbours whose credences differ from their own:

$$P_f(E) = 1 - \min(1,\ d \cdot m)\,(1 - P_i(E)), \tag{3}$$

where $d = |p_i - p_j|$ is the credence distance between receiver and sender, and m is a mistrust multiplier. When $m = 0$ this collapses to Bayes. As $d \cdot m \to 1$, the evidence is discarded entirely.

The simulation framework from *Ignorance in Social Networks* [8] establishes three results:

1. Discounting delays convergence to truth, with statistical significance across all tested parameter combinations (Mann-Whitney U, $p < 0.05$).
2. Network topology affects convergence speed independently of density.
3. When $m > 1$, stable polarisation occurs: the network fractures into subgroups that permanently disagree.

Topology as Allocation Engine

Let $G = (V, E)$ be the analyst network. Shape determines convergence speed and reliability.

Complete Networks

Every agent communicates with every other. Convergence is fast but fragile: early misleading draws propagate everywhere simultaneously, producing false consensus at elevated rates (the Zollman effect [2]). In Rosenstock, Bruner, and O'Connor's extended parameter sweep [5], this fragility persists in the regime of small ε and small N—precisely where real alpha signals operate.

Cycle Networks

Each agent communicates with two neighbours; information propagates around a ring. Convergence is slow but reliable. Misleading evidence is diluted over many hops. The portfolio stays near equal-weight longer, but when it tilts, it tilts correctly with higher probability.

Star Networks

One central hub connects to all peripheral agents, who connect only to it. Convergence is fastest among all tested topologies. The hub anchors expectations. Fragility is concentrated: compromise the hub and the entire network is corrupted. O'Connor and Weatherall note this as a strategy available to disinformants [4].

Scale-Free (Barabási-Albert) Networks

Built by preferential attachment, producing a power-law degree distribution. A few highly-connected hubs dominate information flow; most agents are peripheral. The normalised neighbourhood-size variance (moment_sd) correlates positively with convergence time: heterogeneous information exposure creates stratified belief dynamics.

Density-Controlled Comparison

Networks of size 64 with density ≈ 0.03 produce mean convergence times of 68,818 (star), 109,127 (cycle), 119,976 (Barabási-Albert), and 120,596 (Watts-Strogatz) steps, all pairwise significant at $p < 0.005$ (Mann-Whitney U). Shape, not just connectivity, governs how long ignorance persists. The accompanying simulation[1] replicates these dynamics with the portfolio weight mapping applied.

Risk Aversion as Epistemic Friction

Under Jeffrey updating, the portfolio weight becomes

$$w_B(t) = \frac{1}{2} + \frac{\varepsilon}{2\sigma^2} \cdot \frac{\alpha\, p_i(t)}{\lambda}, \tag{4}$$

where $\alpha \in (0, 1]$ is the effective signal attenuation induced by mistrust. The credence entering the optimiser is not p_i but αp_i. This is algebraically identical to replacing λ with

$$\lambda_{\text{eff}} = \frac{\lambda}{\alpha}. \tag{5}$$

[1] https://leelemacjames.github.io/ignorance-in-portfolios

Epistemic mistrust is isomorphic to increased risk aversion. The topology-dependent consequences:

- In star networks, $\alpha \approx 1$ once convergence begins; $\lambda_{\text{eff}} \approx \lambda$.
- In cycles under Jeffrey updating, α can be substantially less than 1, inflating λ_{eff} and pinning the portfolio near equal-weight.
- In Barabási-Albert networks, α varies by node position, producing a heterogeneous λ_{eff} distribution: agents near hubs rebalance rapidly while peripheral agents remain inert.

Sensitivity

$$\frac{\partial w_B}{\partial \lambda} = -\frac{p_i\,\varepsilon}{2\lambda^2\sigma^2}. \tag{6}$$

Convex in λ: small misspecifications produce disproportionate allocation errors at low λ and diminishing errors at high λ. Since α enters through λ_{eff}, allocation error is a function of three entangled parameters: intrinsic risk aversion, the mistrust multiplier, and network topology.

Polarisation and the Phantom Portfolio

When $m > 1$, one subgroup converges to $p > 0.99$; the remainder stabilise below 0.5, with credence distances large enough that cross-group evidence is discarded. The aggregate portfolio weight

$$\bar{w}_B = \frac{1}{N}\sum_i w_i \tag{7}$$

is held by nobody. It is a statistical artefact of averaging irreconcilable positions. The consensus price it implies does not reflect any agent's belief about ε. This maps to a market split between momentum and value traders occupying non-communicating subnetworks, where the clearing price is set by the capital balance between groups, not by convergent estimation.

This is not a transient disequilibrium. It is a stable fixed point of the belief dynamics.

Endogenous Price Formation

When portfolio weights are translated into aggregate demand, price itself becomes an output of the belief network. In that setting, belief formation and price formation are no longer separable: price is the observable projection of the network's internal epistemic state.

Let price be determined by the aggregate consequence of agent allocations:

$$P(t) = \Phi\big(\{w_i(t)\}_{i=1}^{N}\big), \tag{8}$$

where the weights $w_i(t)$ are themselves functions of credence, mistrust, and topology. Since those weights arise from the interaction of (λ, m, G), price inherits the structure of the network that produces them.

Mispricing is therefore not solely an error relative to fundamentals. It may also be the visible signature of a distorted epistemic process: hub domination, clustered disagreement, or self-referential feedback. In this sense, prices are not merely aggregators of information; they are fixed points of a distributed belief system.

Self-Referential Signals and Coordination

In regimes where exogenous evidence is weak, delayed, or discounted, agents increasingly rely on endogenous signals derived from price and the observed behaviour of others. The update rule takes the form

$$p_i(t+1) = f\big(\{p_j(t)\}_{j \in N(i)}\big), \tag{9}$$

with little or no contribution from independent evidence.

In such cases, commonly observed trading rules based on price patterns function less as evidence about the underlying return process than as coordination devices. Their efficacy depends on shared adoption: they are self-reinforcing rather than truth-tracking. The network may still converge, but the resulting consensus is determined by internal feedback rather than by improved estimation of ε.

This is the limiting case of endogenous market behaviour. Convergence remains possible, but convergence without evidence is not knowledge discovery; it is coordination.

Information Content

A Bayesian agent confronted with a moving average crossover receives no likelihood ratio. The signal conveys what other agents did (in aggregate, with a lag); it conveys nothing about the data-generating process for returns. This is equivalent to setting the number of trials to zero in the simulation. The effective information content about ε is null. The network may converge, but convergence without evidence is convergence without knowledge.

The distinction between rational preference-based allocation and self-fulfilling technical dynamics is precisely the distinction between a network that generates evidence and one that does not.

Toward Endogenous Risk Aversion

If epistemic structure modulates effective risk aversion, λ should not be static. A natural specification:

$$\lambda_i(t) = \lambda_0 + \gamma \cdot \mathrm{Var}_{j \in N(i)}\big(p_j(t)\big), \tag{10}$$

where λ_0 is intrinsic risk aversion and the second term adds a penalty proportional to neighbourhood disagreement. When neighbours agree, the agent invests at their natural aggressiveness. When they disagree, the agent stays closer to equal-weight—a rational response to conflicting information sources.

An alternative links λ directly to evidence quality:

$$\lambda_i(t) = \frac{1}{q_i(t)}, \tag{11}$$

where $q_i(t)$ is the information quality of the agent's received evidence. Both formulations couple portfolio optimisation to belief formation in a way the standard Markowitz framework cannot express.

Empirical Regime Interpretation: FTSE 2008–2023

The endogenous market simulation is calibrated using daily FTSE data from 2008 to 2023, spanning three major episodes of market stress: the Global Financial Crisis (2008–2009), the Brexit referendum (2016), and the COVID-19 pandemic (2020–2021). These episodes are interpreted not simply as exogenous shocks to an otherwise stable market, but as changes in the effective topology and trust structure of the underlying belief network. The purpose of the FTSE

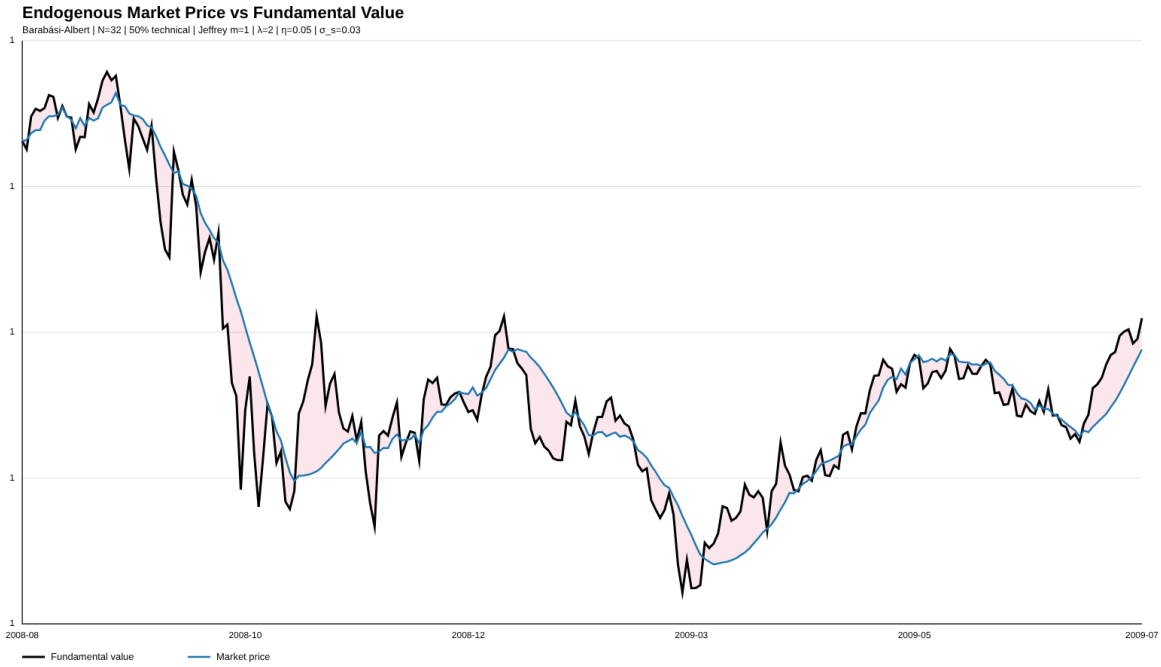

Figure 1: Endogenous market price versus fundamental value (FTSE 100) under a Barabási–Albert network with 32 agents and a 50% share of technical traders. The black line represents observed market fundamentals (FTSE closes), while the blue line shows the simulated endogenous market price generated by networked belief formation.

model is therefore not merely to reproduce volatility, but to illustrate how different epistemic configurations generate distinct price regimes.

The shaded region highlights mispricing, defined as the deviation between endogenous price and fundamental value. This deviation arises not from exogenous shocks alone, but from the internal structure of the belief network, including topology, mistrust, and the presence of self-referential agents.

Even under moderate noise and mixed agent types, the model produces persistent tracking with systematic lag, overshoot, and recovery dynamics. These features are consistent with empirical market behaviour during periods of stress and illustrate how price formation can be understood as the outcome of a distributed epistemic process rather than a direct aggregation of information.

Regime I: Hub Fragility and False Consensus (2008–2009)

The Global Financial Crisis is characterised by the failure of highly centralised information sources: banks, rating agencies, and structured product models. In network terms, this corresponds to a star-like topology in which a small number of hubs dominate information flow.

Under such a structure, convergence is rapid but fragile. Early misleading evidence propagates system-wide before it can be corrected. In the model, this produces high-confidence but incorrect posterior beliefs, leading to coordinated misallocation. The subsequent correction appears as a sharp, discontinuous repricing once the hub is discredited.

Empirically, this is consistent with the observed FTSE trajectory: a rapid, correlated drawdown followed by extreme volatility. In the present framework, this is not merely a reaction to deteriorating fundamentals, but the collapse of a network whose topology amplified initial error.

Regime II: Polarisation and the Phantom Portfolio (2016)

The Brexit referendum introduces a different failure mode. Here, the issue is not a single corrupted hub but persistent disagreement between subgroups interpreting the same evidence differently. Under Jeffrey updating with sufficiently large mistrust parameter m, the network partitions into semi-isolated clusters.

In this regime, belief convergence does not occur globally. Instead, distinct subpopulations stabilise around incompatible credences. The aggregate portfolio weight,

$$\bar{w}_B = \frac{1}{N} \sum_i w_i,$$

is not held by any agent. It is a statistical artefact of averaging across incompatible positions—a phantom portfolio.

The corresponding market price reflects the balance of capital between groups rather than a shared estimate of ε. In the FTSE data, this appears as elevated volatility, abrupt reversals, and sensitivity to political news, consistent with a market in which price is determined by the interaction of non-communicating epistemic communities.

Regime III: Endogenous Feedback and Information Vacuum (2020–2021)

The COVID-19 pandemic represents a third regime: a collapse in the availability of reliable exogenous evidence. At the onset, the distribution of economic outcomes is unknown, and agents lack well-calibrated priors.

In this setting, the belief update rule becomes effectively endogenous. Agents rely increasingly on price signals and the observed behaviour of others. Formally, the update takes the form

$$p_i(t+1) = f\big(\{p_j(t)\}_{j \in N(i)}\big),$$

with negligible contribution from independent evidence. This corresponds to the degenerate epistemology described in Section 4.

Network density increases as attention concentrates on a common set of signals (policy announcements, case counts, market movements), approximating a highly connected graph. Convergence is rapid, but the fixed point reflects internal coordination rather than external truth.

Empirically, this aligns with the FTSE recovery dynamics: a sharp initial decline followed by a rapid rebound, strongly coupled to policy expectations. In the present framework, price is best understood as an emergent consensus of a tightly coupled, self-referential network.

Volatility as a Proxy for Epistemic Dispersion

Equation (10) links effective risk aversion to disagreement within an agent's neighbourhood. In aggregate, dispersion of beliefs manifests as market volatility. Periods of high realised volatility correspond to high variance in $p_i(t)$ across the network, implying elevated λ_{eff} and a collective tendency toward more conservative allocations.

This suggests an interpretation of volatility not merely as a function of exogenous uncertainty, but as a measurable proxy for epistemic disagreement. The FTSE data across all three regimes exhibits pronounced volatility clustering, consistent with persistent disagreement or rapidly shifting network structure.

Synthesis

Across the three episodes, a common pattern emerges. Market behaviour is governed not only by the magnitude of shocks but by the configuration of the belief network through which those shocks are processed. Each regime can be summarised as a distinct configuration of (λ, m, G):

Regime	Topology G	Mistrust m	Outcome
2008–2009	Star-like (hub-dominated)	Low	Fast, fragile convergence; systemic error
2016	Fragmented / clustered	High	Polarisation; phantom portfolio
2020–2021	Dense / near-complete	Moderate	Rapid endogenous convergence

The central implication is that market regimes can be interpreted as phases of a dynamical epistemic system. Price is not simply an aggregator of information but the observable fixed point of a networked belief process whose topology and trust parameters evolve over time.

Conclusion

Portfolio optimisation is not separable from belief formation. The main claim of this paper is that allocation, disagreement, and price formation should all be understood as outputs of a single networked epistemic process.

First, network topology governs the speed and reliability with which agents move from equal-weight ignorance to informed conviction. Star-like structures permit rapid convergence but are vulnerable to systemic error under hub corruption. Cycles and more decentralised structures converge more slowly but are more robust. Scale-free networks produce stratified dynamics in which informational advantage is unevenly distributed across agents.

Second, epistemic mistrust is structurally isomorphic to increased effective risk aversion. The relevant risk parameter observed in behaviour is not intrinsic preference alone, but a composite of preference and epistemic environment:

$$\lambda_{\text{eff}} = \frac{\lambda}{\alpha}.$$

This implies that observed caution in markets may be generated not only by utility curvature, but by disagreement, attenuation, and uncertainty about the reliability of other agents.

Third, when portfolio decisions are aggregated into market demand, price becomes an endogenous object. It reflects the structure of the network through which beliefs are formed and filtered. Mispricing is therefore not solely a deviation from fundamentals; it may also be the visible expression of topological concentration, mistrust, or self-referential feedback.

The FTSE-based endogenous market model:

https://leelemacjames.github.io/endogenous-market.html

illustrates that these regimes are not merely theoretical. The Global Financial Crisis corresponds to hub-dominated fragility; the Brexit period to persistent polarisation; and the COVID-19 period to a dense and highly reflexive market operating under conditions of informational scarcity. Across all three episodes, price formation reflects the epistemic architecture of the market as well as the underlying economic signal.

Allocation error is therefore best understood as a joint function of risk aversion, mistrust, and network geometry:

$$\text{Allocation error} \sim f(\lambda, m, G). \tag{12}$$

More broadly, this suggests a shift in interpretation. Markets are not mechanisms that passively aggregate dispersed information, but dynamical systems in which information is

filtered, transformed, and sometimes endogenously generated. Prices are observable fixed points of a networked belief process, and their informational content depends on the structure of the system that produces them.

In this view, market efficiency is not a property of prices alone, but of the epistemic architecture through which beliefs are formed.

The interactive simulation accompanying this paper is at:

https://leelemacjames.github.io/ignorance-in-portfolios
The endogenous market model is available at:
https://leelemacjames.github.io/endogenous-market.html
Data set from:
https://www.kaggle.com/datasets/pavankrishnanarne/global-stock-market-2008-present
The white paper is hosted at:
https://leelemacjames.github.io/networked-belief-portfolio-allocation.pdf

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